

RESERVED LOCATIONS IN PAGE ZERO

Main memory page zero, between locations 0H and 0FFH, contains several segments of code and data that are used by CP/M:

Locations	Contents
0000H-0002H	Contains a jump instruction to the warm start entry location 4A03H+b. This allows a simple programmed restart (JMP 0000H) or manual restart from the front panel.
0003H-0003H	Contains the Intel standard IOBYTE which is optionally included in the user's CBIOS.
0004H-0004H	Low nibble: current default drive number (0=A,...,15=P) High nibble: current user number.
0005H-0007H	Contains a jump instruction to the BDOS and serves two purposes: JMP 0005H provides the primary entry point to the BDOS, and LHLD 0006H brings the address field of the instruction to the HL register pair. This value is the lowest address in memory used by CP/M, assuming the CCP is being overlaid. The DDT program changes the address field to reflect the reduced memory size in debug mode.
0008H-0027H	Interrupt locations 1 through 5 not used.
0030H-0037H	Interrupt location 6 (not currently used) is reserved.
0038H-003AH	Restart 7; contains a jump instruction into the DDT or SID program when running in debug mode for programmed breakpoints, but is not otherwise used by CP/M.
003BH-003FH	Not currently used; reserved.
0040H-004FH	A 16-byte area reserved for scratch by CBIOS, but is not used for any purpose in the distribution version of CP/M.
0050H-005BH	Not currently used; reserved.
005CH-007CH	Default File Control Block produced for a transient program by the CCP.
007DH-007FH	Optional default random record position.
0080H-00FFH	Default 128-byte disk buffer, also filled with the command line when a transient is loaded under the CCP.

CP/M 2.2 BASIC INPUT/OUTPUT SYSTEM ENTRY POINTS

The BIOS is the machine-dependent part of CP/M. In theory, all you need to do is change the BIOS and CP/M will work on a different machine.

The BIOS begins with the following jumps to service routines:

JMP	BOOT	; -3: Cold start routine
JMP	WBOOT	; 0: Warm boot - reload command processor
JMP	CONST	; 3: Console status
JMP	CONIN	; 6: Console input
JMP	CONOUT	; 9: Console output
JMP	LIST	; 12: Printer output
JMP	PUNCH	; 15: Paper tape punch output
JMP	READER	; 18: Paper tape reader input
JMP	HOME	; 21: Move disc head to track 0
JMP	SELDISK	; 24: Select disc drive
JMP	SETTRK	; 27: Set track number
JMP	SETSEC	; 30: Set sector number
JMP	SETDMA	; 33: Set DMA address
JMP	READ	; 36: Read a sector
JMP	WRITE	; 39: Write a sector
JMP	LISTST	; 42: Status of list device
JMP	SECTTRAN	; 45: Sector translation for skewing

BOOT (FUNCTION 0)

The BOOT entry point gets control from the cold start loader and is responsible for basic system initialization, including sending a sign-on message, which can be omitted in the first version. If the IOBYTE function is implemented, it must be set at this point. The various system parameters that are set by the WBOOT entry point must be initialized, and control is transferred to the CCP at 3400 + b for further processing. Note that register C must be set to zero to select drive A.

WBOOT (FUNCTION 1)

The WBOOT entry point gets control when a warm start occurs. A warm start is performed whenever a user program branches to location 0000H, or when the CPU is reset from the front panel. The CP/M system must be loaded from the first two tracks of drive A up to, but not including, the BIOS, or CBIOS, if the user has completed the patch. System parameters must be initialized as follows:

location 0,1,2: Set to JMP WBOOT for warm starts (000H: JMP 4A03H + b)

location 3: Set initial value of IOBYTE, if implemented in the CBIOS

location 4: High nibble = current user number, low nibble = current drive

location 5,6,7: Set to JMP BDOS, which is the primary entry point to CP/M for transient programs. (0005H: JMP 3C06H + b)

CONST (FUNCTION 2)

Returns its status in A; 0 if no console character is ready, 0FFh if one is.

CONIN (FUNCTION 3)

Wait until the console keyboard is ready to provide a character, and return it in A.

CONOUT (FUNCTION 4)

Write the character in C to the console.

LIST (FUNCTION 5)

Write the character in C to the printer. If the printer isn't ready, wait until it is.

PUNCH / AUXOUT (FUNCTION 6)

Write the character in C to the "paper tape punch" - or whatever the current auxiliary device is. If the device isn't ready, wait until it is.

READER (FUNCTION 7)

Read a character from the "paper tape reader" - or whatever the current auxiliary device is. If the device isn't ready, wait until it is. The character will be returned in A. If this device isn't implemented, return character 26 (^Z).

HOME (FUNCTION 8)

The disk head of the currently selected disk (initially disk A) is moved to the track 00 position. If the controller allows access to the track 0 flag from the drive, the head is stepped until the track 0 flag is detected. If the controller does not support this feature, the HOME call is translated into a call to SETTRK with a parameter of 0.

SELDSK (FUNCTION 9)

Select the disc drive in register C (0=A:, 1=B: ...). Called with E=0 or 0FFh.

If bit 0 of E is 0, then the disk is logged in as if new; if the format has to be determined from the boot sector, for example, this will be done.

If bit 0 of E is 1, then the disk has been logged in before. The disc is not accessed; the [DPH](#) address (or zero) is returned immediately.

The disk drive given by register C is selected for further operations, where register C contains 0 for drive A, 1 for drive B, and so on up to 15 for drive P (the standard CP/M distribution version supports four drives). On each disk select, SELDSK must return in HL the base address of a 16-byte Disk Parameter Header. For standard floppy disk drives, the contents of the header and associated tables do not change; thus, the program segment included in the sample CBIOS performs this operation automatically.

If there is an attempt to select a nonexistent drive, SELDSK returns HL = 0000H as an error indicator.

Although SELDSK must return the header address on each call, it is advisable to postpone the physical disk select operation until an I/O function (seek, read, or write) is actually performed, because disk selects often occur without ultimately performing any disk I/O, and many controllers unload the head of the current disk before selecting the new drive. This causes an excessive amount of noise and disk wear.

The least significant bit of register E is zero if this is the first occurrence of the drive select since the last cold or warm start.

SETTRK (FUNCTION 10)

Register BC contains the track number for subsequent disk accesses on the currently selected drive. The sector number in BC is the same as the number returned from the SECTRA entry point. You can choose to seek the selected track at this time or delay the seek until the next read or write actually occurs. Register BC can take on values in the range 0-76 corresponding to valid track numbers for standard floppy disk drives and 0- 65535 for nonstandard disk subsystems.

SETSEC (FUNCTION 11)

Register BC contains the sector number, 1 through 26, for subsequent disk accesses on the currently selected drive. The sector number in BC is the same as the number returned from the SECTRA entry point. You can choose to send this information to the controller at this point or delay sector selection until a read or write operation occurs.

SETDMA (FUNCTION 12)

Register BC contains the DMA (Disk Memory Access) address for subsequent read or write operations. For example, if B = 00H and C = 80H when SETDMA is called, all subsequent read operations read their data into 80H through 0FFH and all subsequent write operations get their data from 80H through 0FFH, until the next call to SETDMA occurs. The initial DMA address is assumed to be 80H. The controller need not actually support Direct Memory Access. If, for example, all data transfers are through I/O ports, the CBIOS that is constructed uses the 128 byte area starting at the selected DMA address for the memory buffer during the subsequent read or write operations.

READ (FUNCTION 13)

Assuming the drive has been selected, the track has been set, and the DMA address has been specified, the READ subroutine attempts to read one sector based upon these parameters and returns the following error codes in register A:

0 - no errors occurred

1 - nonrecoverable error condition occurred

Currently, CP/M responds only to a zero or nonzero value as the return code. That is, if the value in register A is 0, CP/M assumes that the disk operation was completed properly. If an error occurs the CBIOS should attempt at least 10 retries to see if the error is recoverable. When an error is reported the BDOS prints the message BDOS ERR ON x: BAD SECTOR. The operator then has the option of pressing a carriage return to ignore the error, or CTRL-C to abort.

WRITE (FUNCTION 14)

EData is written from the currently selected DMA address to the currently selected drive, track, and sector. C contains a deblocking code:

C=0 - Write can be deferred

C=1 - Write must be immediate

C=2 - Write can be deferred, no pre-read is necessary.

Returns A=0 for OK, 1 for unrecoverable error, 2 if disc is read-only, 0FFh if media changed.

LISTST (FUNCTION 15)

Return status of current printer device.

Returns A=0 (not ready) or A=0FFh (ready).

SECTTRAN (FUNCTION 16)

Translate sector numbers to take account skewing.

On entry, BC=logical sector number (zero based) and DE=address of translation table. On exit, HL contains physical sector number. On a system with hardware skewing, this would normally ignore DE and return either BC or BC+1.

Logical-to-physical sector translation is performed to improve the overall response of CP/M. Standard CP/M systems shipped with a skew factor of 6, where six physical sectors are skipped between each logical read operation. (This skew factor allows enough time between sectors for most programs to load their buffers without missing the next sector.)

In general, SECTTRAN receives a logical sector number relative to zero in BC and a translate table address in DE. The sector number is used as an index into the translate table, with the resulting physical sector number in HL. For standard systems, the table and indexing code is provided in the CBIOS and need not be changed.

CP/M 2.2 DISC PARAMETER HEADER

In CP/M, the DPH is a BIOS structure containing information about a disc drive. (There is one DPH per disk drive.) Given n disk drives, the DPHs are arranged in an array that is $n \times 16$ bytes long. The first row of 16 bytes corresponds to drive 0, with the last row corresponding to drive $n-1$. (Individual BIOSes may add extra fields to a DPH, before the start or after the end.)

The address of the DPH depends on the machine used, but it can be determined by a simple BIOS-call (function 9: SELECT DISK).

DPH Offset	Contents	Meaning	
\$00	DPBASE	Base address of the DPH table.	
\$00	XLT	Address of the logical-to-physical sector translation table, if used for this particular drive. Otherwise, the value of \$0000 if there is no there is no translation table for this drive (that is, the physical and logical sector numbers are the same). Disk drives with identical sector translation can share the same translate table.	
\$02	TRACK	BDOS work area for the current track number	Three scratch pad words for use within the BDOS. The initial value is unimportant.
\$04	SECTOR	BDOS work area for the current sector number	
\$06	DIRNUM	BDOS work area for the current directory number	
\$08	DIRBUF	Address of a 128-byte scratch pad area (directory buffer) for directory operations within BDOS. All DPHs address the same scratch pad area.	
\$0A	DPB	Address of a Disk Parameter Block for this drive. Drives with identical disk characteristics can address the same disk parameter block.	
\$0C	CSV	Check Sum Vector: Holds the checksum of each 128-byte record in the directory. If the checksum of a sector changes unexpectedly, the disc is assumed to have been changed. If the disc is not removable, then there will be no such buffer, and its address in the DPH is 0. This address is different for each DPH.	
\$0E	ALV	Allocation Vector: Address of a scratch pad area used by the BDOS to keep disk storage allocation information. This is a bitmap of used/free blocks on the drive. Under CP/M 2, there is one bit per block. This address is different for each DPH.	
\$0F		last byte of ALV and the DPH	

SELDSK

A responsibility of the SELDSK subroutine is to return the base address of the DPH for the selected drive. The following sequence of operations returns the table address, with a \$0000 returned if the selected drive does not exit.

SELDISK example:

```
NDISKS    EQU    4                ;Number of disk drives
.....
SELDISK:   ;Select disk given by BC
           LXI    H,$0000          ;Error code
           MOV    A,C              ;drive ok?
           CPI    NDISKS           ;CY IF SO
           RNC                      ;RET f error

           ;no error, continue
           MOV    L,C              ;low (disk)
           MOV    H,B              ;high (disk)
           DAD    H                ;*2
           DAD    H                ;*4
           DAD    H                ;*8
           DAD    H                ;*16
           LXI    D,DBASE          ;first DPH
           DAD    D                ;DPH (disk)
           RET
```

The translation vectors XLT are located elsewhere in the BIOS, and simply correspond one-for-one with the logical sector numbers zero through the sector count 1.

The pointer to the DPB is located in a DPH at offset \$0A. Note, that CP/M manages mass storage devices in 128-bytes sectors, regardless how many bytes are in the physical sectors of a disk. Several of such sectors are combined to "Allocation Blocks". The Allocation Blocks are combined to "Logical Extents" for the representation in the directory.

DISK PARAMETER BLOCK

The Disk Parameter Block (DPB) summarizes the attributes of identical groups of mass storage devices.

The 15-bytes DPB describes the attributes of a drive and has this general form:

DPB Offset	Contents	Meaning
\$00	SPT	Sectors Per Track: Total number of 128 bytes sectors per track
\$02	BSH	Data allocation Block Shift Factor: Number of 128 bytes sectors per "Allocation Block" (determined by the data block allocation size. Stored as 2's-logarithm. See notes below).
\$03	BLM	Data allocation Block Mask: To get the sector index of the Allocation Block the sector number is to combine logically by AND with this mask. ($(2^{BSH}-1)$). See notes below.
\$04	EXM	Extent Mask: Number of extents per directory entry (stored as 2's logarithm). Determined by the data block allocation size and the number of disk blocks.
\$05	DSM	Total storage capacity of the disk drive. Number of the last Allocation Block (Number of blocks per disk -1)
\$07	DRM	Number of the last directory entry (Number of directory entries -1). Total number of directory entries that can be stored on this drive. (AL0, AL1 determine reserved directory blocks).
\$09	AL0	Starting value of the first two bytes of the allocation table. (Determined by DRM; bit meaning see below)
\$0A	AL1	
\$0B	CKS	Check area Size: Size of the directory check vector. Number of directory entries to check for disk change. Zero for a HDD.
\$0D	OFF	Number of system reserved tracks at the beginning of the (logical) disk

The values of BSH and BLM implicitly determine the data allocation size, BLS, which is not an entry in the DPB.

Given that the designer has selected a value for BLS, the values of BSH and BLM are shown in the following table:

BLS	BSH	BLM
1,024	3	7
2,048	4	15
4,096	5	31
8,192	6	63
16,384	7	127

All values are decimal. The value of EXM depends upon the BLS and whether the DSM value is less than 256 or greater than 255. For DSM less than 256, the value of EXM is given by:

4	EXM
1,024	0
2,048	1
4,096	3
8,192	7
16,384	15

For DSM greater than 255, the value of EXM is given by:

BLS	EXM
1,024	N/A
2,048	0
4,096	1
8,192	3
16,384	7

The value of DSM is the maximum data block number measured in BLS units supported by this particular drive. The product $BLS * (DSM + 1)$ is the total number of bytes held by the drive and, of course, must be within the capacity of the physical disk, not counting the reserved operating system tracks.

The DRM entry is one less than the total number of directory entries that can take on a 16-bit value. The values of AL0 and AL1 are determined by DRM. AL0 and AL1 values together can be considered a string of 16-bits, as shown below:

AL0								AL1							
00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15

Position 00 corresponds to the high-order bit of the byte AL0, and position 15 to the low-order bit of the byte AL1. Each bit position reserves a data block for a number of directory entries, thus allowing a total of 16 data blocks to be assigned for directory entries (bits are assigned starting at 00 and filled to the right through position 15). Each directory entry occupies 32 bytes, resulting in the following table:

BLS	Directory Entries
1,024	32 times # bits
2,048	64 times # bits
4,096	128 times # bits
8,192	256 times # bits
16,384	512 times # bits

If $DRM = 127$ (128 directory entries) and $BLS = 1,024$, there are 32 directory entries per block, requiring four reserved blocks. In this case, the four high-order bits of AL0 are set, resulting in the values $AL0 = \$F0$ and $AL1 = \$00$.

The CKS value is determined as follows:

If the disk drive media is removable, then $CKS = (DRM + 1) / 4$, where DRM is the last directory number.

If the media are fixed, then $CKS = 0$. No directory records are checked in this case.

Finally, the OFF field determines the number of tracks that are skipped at the beginning of the physical disk. This value is automatically added whenever SETTRK is called. It can be used as a mechanism for skipping reserved operating system tracks or for partitioning a large disk into smaller segmented sections.

To complete the discussion of the DPB, several DPHs can address the same DPB if their drive characteristics are identical. Further, the DPB can be dynamically changed when a new drive is addressed. Since the BDOS copies the DPB values to a local area whenever the SELDSK function is called, simply change the pointer in the DPH.

Returning back to the DPH for a particular drive, the two address values, CSV and ALV, reference areas of uninitialized memory in the BIOS data segment. The areas must be unique for each drive, and the size of each area is determined by the values in the DPB.

The size of the data addressed by CSV is CKS bytes, which is sufficient to hold the directory check information for this particular drive. If $CKS = (DRM + 1) / 4$, you must reserve $(DRM + 1) / 4$ bytes for directory check use. If $CKS = 0$, no storage is reserved.

The size of the area addressed by ALV is determined by the maximum number of data blocks allowed for this particular disk, and is equal to $2 * (DSM / 8 + 1)$. Two copies of the allocation map for the disk are kept in this area: the first vector stores temporarily allocated blocks resulting from write operations, the second stores permanently allocated blocks resulting from CLOSE FILE operations.

The DPB is not stored on disc. It is either hardwired into the BIOS or generated on the fly when a disc is logged in.

The reserved tracks will contain an image of CP/M 2.2, used when the system is rebooted. Discs can be formatted as data only discs, in which case they have no system tracks and cannot be used to reboot the system.

DISK ORGANIZATION

The sector allocation for the standard distribution version of CP/M is given here for reference purposes. The first sector contains an optional software boot section (see the table on the following page). Disk controllers are often set up to bring track 0, sector 1, into memory at a specific location, often location 0000H. The program in this sector, called BOOT, has the responsibility of bringing the remaining sectors into memory starting at location 3400H + b. If the controller does not have a built-in sector load, the program in track 0, sector 1 can be ignored. In this case, load the program from track 0, sector 2, to location 3400H + b.

As an example, the Intel MDS-800 hardware cold start loader brings track 0, sector 1, into absolute address 3000H. Upon loading this sector, control transfers to location 3000H, where the bootstrap operation commences by loading the remainder of track 0 and all of track 1 into memory, starting at 3400H + b. Note that this bootstrap loader is of little use in a non-MDS environment, although it is useful to examine it because some of the boot actions will have to be duplicated in the user's cold start loader.

Track	Sector	Memory Address	CP/M Module name
00	01	(boot address)	Cold Start Loader
00	02	3400H + b	CCP
	03	3480H + b	
	04	3500H + b	
	05	3580H + b	
	06	3600H + b	
	07	3680H + b	
	08	3700H + b	
	09	3780H + b	
	10	3800H + b	
	11	3880H + b	
	12	3900H + b	
	13	3980H + b	
	14	3A00H + b	
	15	3A80H + b	
	16	3B00H + b	
	17	3B80H + b	
00	18	3C00H + b	BDOS
	19	3C80H + b	
	20	3D00H + b	
	21	3D80H + b	
	22	3E00H + b	
	23	3E80H + b	
	24	3F00H + b	
	25	3F80H + b	
	26	4000H + b	
01	01	4080H + b	
	02	4100H + b	
	03	4180H + b	
	04	4200H + b	
	05	4280H + b	
	06	4300H + b	
	07	4380H + b	
	08	4400H + b	
	09	4480H + b	
	10	4500H + b	
	11	4580H + b	
	12	4600H + b	
	13	4680H + b	
	14	4700H + b	
	15	4780H + b	
	16	4800H + b	
	17	4880H + b	
	18	4900H + b	
	19	4980H + b	
01	20	4A00H + b	BIOS
	21	4A80H + b	
	22	4B00H + b	
	23	4B80H + b	
	24	4C00H + b	
	25	4C80H + b	
	26	4D00H + b	

CP/M 2.2 DIRECTORY

The CP/M 2.2 directory has only one type of entry:

```
UU F1 F2 F3 F4 F5 F6 F7 F8 T1 T2 T3 EX S1 S2 RC   .FILENAME.TYP....
AL AL AL AL AL AL AL AL AL AL AL AL AL AL AL AL   .....
```

UU = User number. 0-15 (on some systems, 0-31). The user number allows multiple files of the same name to coexist on the disc.

User number = 0E5h => File deleted

Fn = filename

Tn = filetype. The characters used for these are 7-bit ASCII.

The top bit of T1 (often referred to as T1') is set if the file is read-only.

T2' is set if the file is a system file (this corresponds to "hidden" on other systems).

EX = Extent counter, low byte - takes values from 0-31

S2 = Extent counter, high byte.

An extent is the portion of a file controlled by one directory entry. If a file takes up more blocks than can be listed in one directory entry, then it is given multiple entries, distinguished by their EX and S2 bytes. The formula is: Entry number = $((32 * S2) + EX) / (exm + 1)$ where exm is the extent mask value from the Disc Parameter Block.

S1 = reserved, set to 0.

RC = Number of records (1 record=128 bytes) used in this extent, low byte.

The total number of records used in this extent is

$$(EX \& exm) * 128 + RC$$

If RC is 80h, this extent is full and there may be another one on the disc.

File lengths are only saved to the nearest 128 bytes.

AL = Allocation. Each AL is the number of a block on the disc. If an AL number is zero, that section of the file has no storage allocated to it (ie it does not exist). For example, a 3k file might have allocation 5,6,8,0,0....- the first 1k is in block 5, the second in block 6, the third in block 8.

AL numbers can either be 8-bit (if there are fewer than 256 blocks on the disc) or 16-bit (stored low byte first).